a data structure that uses a tiny variation on the approach in section 8.4.3 to avoid the exponential growth of the tree.

8.5 Summary

In this chapter we have moved from unidimensional datasets to multidimensional data, reasoning on the issues and possible solutions that this shift causes. A few of the most relevant points tackled in this chapter include:

- We have introduced a key problem in modern data science, finding the nearest neighbor(s) of a given entry.
- We can't use a conventional container to handle multidimensional data.
- When the number of dimensions of a dataset grows, indexing with traditional methods become unfeasible due to the exponential growth of the number of branches at each step.
- We defined a common API for a class of containers holding multi-dimensional data, providing a method to query for the closest points to an arbitrary target.

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[1] Here rotation refers to the balancing operation performed, among others, on red-black trees.

[2] An Euclidian distance is the ordinary straight-line distance between two points in Euclidean space, like the Euclidean plane or 3-D Euclidean space, and their generalizations to k dimensions.

[3] An octant is, by definition, one of the eight divisions of a Euclidean three-dimensional coordinate system. Usually octants refer to the 8 hyper-cubes resulting from splitting $\mathbb{R}^3$ along the three cartesian axes, so each octant is defined by the sign of the coordinates: for